SEMAPHORES

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Write a C program to implement the **Producer–Consumer problem** using semaphores. The program should simulate a bounded buffer where a producer produces items and a consumer consumes items.

The program must use the following variables:

- `mutex` → to ensure mutual exclusion
- `full` → to count filled slots
- `empty` → to count empty slots
- `x` → to track number of items

The user should be able to choose:

1. Producer
2. Consumer
3. Exit

Sample Input 1:

1

1

2

3

Sample Output 1:

----- Producer Consumer Problem -----

1. **Producer**
2. **Consumer**
3. **Exit**

Enter your choice: 1 Producer produces item 1

Enter your choice: 1 Producer produces item 2

Enter your choice: 2 Consumer consumes item 2

Enter your choice: 3 Exiting program... Final buffer items produced: 1

Sample Input 2:

2

1

3

Sample Output 2:

----- Producer Consumer Problem -----

Enter your choice: 2 Buffer is empty!! Consumer cannot consume.

Enter your choice: 1 Producer produces item 1

Enter your choice: 3 Exiting program... Final buffer items produced: 1

CODE:

```
#include <stdio.h>

#include <stdlib.h>

int mutex = 1, full = 0, empty = 5, x = 0;

int wait(int s)
{

    return --s;

}

int signal(int s)
```

```
{
    return ++s;
}
void producer()
{
    mutex = wait(mutex);
    empty = wait(empty);
    full = signal(full);
    x++;
    printf("\nProducer produces item %d", x);
    mutex = signal(mutex);
}
void consumer()
{
    mutex = wait(mutex);
    full = wait(full);
    empty = signal(empty);
    printf("\nConsumer consumes item %d", x);
    x--;
    mutex = signal(mutex);
}
```

```
int main()

{    int
n;

   printf("\n----- Producer Consumer Problem -----");
printf("\n1. Producer");     printf("\n2. Consumer");
printf("\n3. Exit");


   while(1)

   {

       printf("\nEnter your choice: ");
scanf("%d",&n);

       switch(n)

       {

          case 1:

             if(mutex==1 && empty!=0)

             {

                producer();

             }
else

             {

                printf("\nBuffer is full!! Producer cannot produce.");

             }

             break;

          case 2:
```

```
            if(mutex==1 && full!=0)

            {
               consumer();

            }
else

            {

               printf("\nBuffer is empty!! Consumer cannot consume.");

            }

            break;

         case 3:

            printf("\nExiting program...");
printf("\nFinal buffer items produced: %d\n", x);

            exit(0);



         default:

            printf("\nInvalid choice. Try again.");

      }

   }



   return 0;

}
```